Here is Puppet Your Configuration Manager

Puppet is an open source configuration management utility that runs on several UNIX-like systems as well as Microsoft Windows. It helps sysadmins automate many repetitive tasks.

A group of computers in a network is called a cluster. In many organisations, you will find the server being set up in a cluster so that if one machine fails, the process goes on. Even in Hadoop, the data is scattered over the cluster. Maintenance of these clusters is a cumbersome job for the administrator and developers. For example, let's assume that in a cluster of 1000 nodes (nodes refer to physical machines), there is a software update that needs to be applied on all the machines. Now, if you don't have a configuration management system, then you have to manually install the software on all the machines. This is really tiring when you have many software updates or configuration changes on a frequent basis.

To ease the job, configuration management systems were created and Puppet is one of them. It allows us to define the state of our IT infrastructure, and then automatically enforces the desired state. Puppet automates every step of the software delivery process, from provisioning of physical and virtual machines to orchestration and reporting; from early-stage code development through testing, production release and updates. Puppet has two solutions - one for configuration management and the other for data centre automation. In this article, we will look at its configuration management feature.

Puppet works in a master-slave design, where all the required configuration is done on the master and is reflected in all the slaves on a regular basis. The slave machine syncs up with the master every 30 minutes to reflect the configuration of the master. I will tell you how to: a) Install the latest version of Ruby; b) Install and configure a central Puppet master server; c) Install and configure a Puppet agent node; d) Sign the agent's SSL certificate request on the master, and finally; e) Execute a simple test with Puppet.

Ruby is one of the basic requirements for installing and configuring the Puppet agent/master set-up. So let's install it in on both master and slave. Use one of the following versions of MRI (standard) Ruby: 2.1.x, 2.0.x, 1.9.3 or 1.8.7.

Steps for installing Ruby

1. We first need to install all the required packages for Ruby installation on our system using the following commands:

```
yum install gcc-c++ patch readline readline-devel zlib zlib-devel
yum install libyaml-devel libffi-devel openssl-devel make
yum install bzip2 autoconf automake libtool bison iconv-devel
```